

INFO 5602
Information Visualization

“Bad Graphs Are Everywhere”

Agenda

- Bad graphs (presentations) are everywhere
- We are NOT naturally good at storytelling with data
- Anyone can learn to use Excel – but not everyone understands the intricacies of data/information visualization

Bad graphs are everywhere!

- What is a bad graph (class/group discussion)?
- Nobody sets out to make a bad graph/presentation - but it happens!
- Bad graphs have become prevalent in the media and in places you expect people to know better – because you think they should have been **trained** to know better
- A combination of **bad graphs** and **poor storytelling** can negatively affect our ability to achieve effective communication with our stakeholders
 - Movies that win awards are mostly storied told in a manner that effectively communicates with stakeholders (movie goers, who say yes or no to tickets)

We are NOT naturally good at **storytelling**

- We are good at some subjects (e.g. math etc.) because we have been formally trained overtime on the rules of math
- You need to be a good storyteller to convince people to do something
- How many have been formally trained in the science and art of storytelling?
- Many of us focus on our own needs and view of the world when trying to convince others to make decisions or take actions
- We tend to **assume** stakeholders/audiences have time, want to listen to what we have to say/present and **we rarely think about attention deficiency when we encounter them**
 - Remember: its not about you; its about your audience/stakeholder(s)

We are NOT naturally good at **storytelling**

- In addition to a lack of formal, long-term training in storytelling other challenges abound:
 - We are not formally trained in telling stories with numbers/datasets
 - Few people feel naturally adept with telling stories with numbers
 - Stage fright is an ongoing issue when you have to present to stakeholders
- Storytelling with numbers/datasets is high demand in many industries because of big data (volume, variety, velocity) and technology advances
- Being able to visualize data and tell stories with it is key to achieving effective stakeholder communication – and better decision-making
- Storytelling can be learned as a skill to help stakeholders retain information for effective decision-making
 - Body language + trust building
 - **Repetition** helps attention-deficient stakeholders retain information
 - Proper **presentation structuring**

We are NOT naturally good at **storytelling** and **presentation structuring**

Presentation = Noise + Signal

- *Clutter/distractions on slides*
- *Stakeholder - irrelevant information included*
- *Stakeholder -irrelevant “design” options*
- *Distracting colors, objects included in presentations*
- **Adds cognitive load to stakeholders**

Find + Remove

- *Stakeholder-relevant information*
- *Stakeholder-relevant charts*
- *Stakeholder-relevant design options*
- *User-centric design for charts*
- **Improves effective stakeholder communication**

Find/Create + Boost

Anything you add to this *blank* slide adds
cognitive load to an audience

Anyone can use Excel etc. to create a graph

- We are not very good at telling stories with numbers – so we hide behind the tools (Excel, Tableau, PowerBI etc) and hope **flashy charts will save us**
- Anyone can watch YouTube and learn how to use Excel and other tools to convert numbers to charts.
 - There is no guarantee that our charts will matter to stakeholder regardless of how “flashy” or “pretty” or “techy” **those charts are made to look**
- Stakeholders will care about our charts to the extent **we care about them and what they care about**. Its not about us, its about our stakeholders
- Key takeaway: you need to enhance your tool skills with **formal training in how to tell stories with data/information visualization**

End of presentation